Introduction to Databases

- A database is a structured way to store and manage data.
- It helps organize data for easy access, retrieval, and updates.
- Databases store data like tables, documents, or key-value pairs.
- They are used in applications to handle user and system data.
- Databases ensure data is consistent and available when needed.
- Examples include MySQL, MongoDB, PostgreSQL, and Redis.

Why Do We Need Databases?

- **Data Integrity:** Ensures data consistency. For example, in banking, a transaction either happens fully or not at all.
- **Concurrency Management:** Discuss how multiple users can access the same data simultaneously without conflicts.
- **Backup & Recovery:** Automatic processes to save data and restore it after failures.
- **Scalability:** Mention how databases are designed to handle growth—from hundreds to millions of records.

Types of Databases

SQL Databases	NoSQL Databases
Store data in **structured tables** with **rows and columns**.	Designed for **flexibility** with **unstructured, semi-structured**, or large-scale data.
Use **predefined schemas**, where the structure is fixed before data entry.	**Do not require a fixed schema**, making them ideal for dynamic data models.
Support complex queries using **SQL (Structured Query Language).**	Support complex queries using query languages or APIs specific to each NoSQL database, such as MongoDB's JSON-like queries, Cassandra's CQL (Cassandra Query Language), or Neo4j's Cypher for graph traversal.
Examples: MySQL, PostgreSQL, SQL Server, etc.	Examples based on format in which data is stored: • **Key-Value**: Redis, DynamoDB (e.g., caching or session data). • **Document**: MongoDB, CouchDB (e.g., JSON-like documents for user profiles). • **Graph**: Neo4j, ArangoDB (e.g., social networks or recommendation engines).